Andy Amponsah

DevOps Engineer | Cloud Solutions Specialist

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Professional Summary

Certified DevOps Engineer with 3+ years of experience designing, implementing, and optimizing cloud infrastructure and CI/CD pipelines. AWS Certified Cloud Practitioner with expertise in Infrastructure as Code (Terraform, CloudFormation), containerization (Docker, Kubernetes), and automation tools (Jenkins, GitHub Actions). Former full-stack developer with strong Python and Django skills, bringing a development perspective to operational challenges. Proven ability to reduce deployment times by 40% while improving system reliability. Passionate about implementing DevOps best practices, security protocols, and cost-effective cloud solutions.

Professional Experience

DevOps Engineer | AmaliTech — April 2024 - Present
Kumasi, Ghana

- Designed and implemented CI/CD pipelines using Jenkins, reducing deployment times by 40% through parallel job execution and optimized build processes
- Automated infrastructure provisioning using Terraform and AWS CloudFormation, managing 50+ EC2 instances, RDS databases, and S3 buckets across 3 environments
- Implemented comprehensive monitoring solutions with CloudWatch and Prometheus/Grafana, reducing incident response time by 60% through proactive alerting
- Containerized 15+ applications using Docker and orchestrated deployments with Kubernetes (EKS), improving scalability and resource utilization
- Established DevOps best practices including infrastructure as code, immutable deployments, and blue-green deployment strategies
- Reduced AWS costs by 25% through rightsizing instances, implementing auto-scaling, and optimizing storage solutions

AWS | Terraform | Jenkins CI/CD | Docker | Kubernetes | Infrastructure as Code

Cloud Engineering Fellow | Azubi Africa Talent Mobility Program — July 2025 - Present
Remote

- Essential cloud engineering training program with focus on AWS services, Linux administration, and DevOps methodologies
- Designed and deployed highly available cloud architecture for web applications using AWS services (EC2, S3, RDS, Lambda)
- Implemented automated backup solutions for critical data using AWS Backup and lifecycle policies
- Collaborated with cross-functional teams on migration of legacy systems to cloud infrastructure
- Developed security best practices including IAM policies, security groups, and encryption standards

AWS Cloud | Linux Administration | Networking | Security | Python Scripting

Founder & Lead Developer | Kryotech Software Solutions — March 2018 - Present
Kumasi, Ghana (Remote)

- Architected, developed and deployed 15+ Django web applications with automated CI/CD pipelines using GitHub Actions and Jenkins
- Implemented comprehensive automated testing frameworks (unit, integration, E2E) reducing production bugs by 35%
- Migrated 8 legacy systems to AWS cloud infrastructure, improving scalability and reducing downtime by 90%
- Led agile development team of 3 developers, implementing Scrum methodologies that improved delivery time by 25%
- Provided cybersecurity consulting for SMBs, implementing security measures that prevented 100% of attempted breaches

Django | Python | AWS Migration | CI/CD | Cybersecurity | Team Leadership

Coding Instructor | National Service Secretariat — March 2024 - April 2025
Kumasi, Ghana

- Developed and delivered comprehensive curriculum for Python programming fundamentals and web development
- Taught full-stack development using HTML5, CSS3, JavaScript, and Django framework to 50+ students
- Mentored students in software development best practices including version control, testing, and documentation

Python | Web Development | Curriculum Development | Mentoring

Education & Training

AWS re/Start Program - Cloud Computing — 2024
AmaliTech Ghana
Intensive 12-week cloud computing program covering AWS fundamentals, Linux administration, networking, security, Python scripting, and automation. Completed hands-on labs and real-world projects.

Bachelor of Arts in Social Studies & History — 2019 - 2023
University of Ghana
Combined Major, GPA: 3.55/4.0. Relevant coursework: Research Methods, Data Analysis, Critical Thinking

Google IT Support Professional Certificate — 2021
Google (Coursera)
Covered troubleshooting, customer service, networking, operating systems, system administration, and security

CS150 Introduction to Computer Science — 2021
Harvard (Online)
A comprehensive course covering fundermentals of computer science, networking, security and data structures

Azubi Africa Talent Mobility Program - Cloud Engineering — 2023 - 2024
Azubi Africa
Intensive training in cloud computing, DevOps practices, and AWS services. Focused on practical implementation of scalable cloud solutions.

Certifications

AWS Certified Cloud Practitioner
Amazon Web Services
Validation Number: [Your Validation Number] | Issued 2024

Google IT Support Specialist
Google
Credential ID: [Your Credential ID] | Issued 2021

Python for Everybody Specialization
University of Michigan (Coursera)
Credential ID: [Your Credential ID] | Issued 2021

Azubi Africa Cloud Engineering Certificate
Azubi Africa
Issued 2024

Technical Skills

Cloud & DevOps
- AWS (EC2, S3, Lambda, RDS, IAM, VPC, CloudFront)
- Infrastructure as Code (Terraform, CloudFormation)
- CI/CD Pipelines (Jenkins, GitHub Actions, CircleCI)
- Containerization (Docker, Kubernetes, ECS, EKS)
- Configuration Management (Ansible, Chef)

Programming & Scripting
- Python (Django, Flask, Pandas, NumPy)
- Bash/Shell Scripting (Automation, Cron Jobs)
- JavaScript (Node.js, Express)
- SQL (PostgreSQL, MySQL) & NoSQL (MongoDB)
- YAML, JSON (Configuration Files)

Other Technologies
- Linux System Administration (Ubuntu, CentOS)
- Monitoring (Prometheus, Grafana, CloudWatch)
- Version Control (Git, GitHub, GitLab, Bitbucket)
- Agile & Scrum Methodologies
- Networking & Security (TCP/IP, Firewalls, VPN)

Notable Projects

Automated Cloud Infrastructure Provisioning System
Terraform-based solution for rapid environment setup across AWS

- Developed modular Terraform templates for VPCs, EC2, RDS, S3, and security groups
- Reduced infrastructure setup time from 8 hours to 15 minutes for new environments
- Implemented state management with S3 backend and locking via DynamoDB
- Integrated with AWS Secrets Manager for secure credential management
- GitHub: https://github.com/guderian120/terraform-aws-infrastructure

Kubernetes Cluster for Microservices Architecture
EKS implementation for containerized applications

- Designed and deployed production-grade EKS cluster with 10+ microservices
- Implemented horizontal pod autoscaling based on custom metrics
- Configured Istio service mesh for advanced traffic management and observability
- Reduced infrastructure costs by 30% through efficient resource utilization
- GitHub: https://github.com/guderian120/kubernetes-microservices

Jenkins CI/CD Pipeline for Django Applications
End-to-end automation from commit to production

- Built multi-stage pipeline with parallel test execution
- Integrated SonarQube for static code analysis with quality gates
- Implemented blue-green deployment strategy with zero downtime
- Reduced deployment time from 15 minutes to under 2 minutes
- GitHub: https://github.com/guderian120/jenkins-django-pipeline

Cloud WorkSpace Provisioning System
Automated user management for cloud environments

- Built using Django, Pandas, and hosted on AWS EC2
- Integrated GitHub Actions for full CI/CD automation including testing and deployment
- Enabled sysadmins to provision/deprovision user workspaces with appropriate permissions
- Reduced onboarding time for new developers from 2 days to 30 minutes
- Live Demo: https://github.com/guderian120/Bash_User_management_System

Peer Review Hub Dashboard
Interactive dashboard for tracking code review metrics

- Built with HTML5, CSS3, JavaScript, and Chart.js for visualizations
- Hosted on GitHub Pages with automated updates via GitHub Actions
- Tracks review turnaround time, feedback quality, and improvement trends
- Helped improve code review efficiency by 40% through data-driven insights
- Live App: https://guderian120.github.io/peer_reviews/index.html

Additional Information

Languages
- English (Fluent)
- German (Beginner)

Professional Affiliations
- Member, AWS User Group Ghana
- Member, DevOps Ghana Community